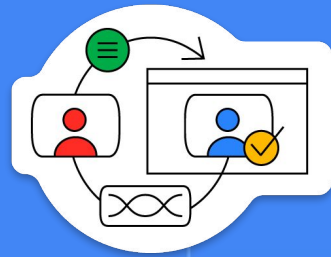


Intro to GAI



Imen Selmi

AI Engineer / WTM





CHATGPT



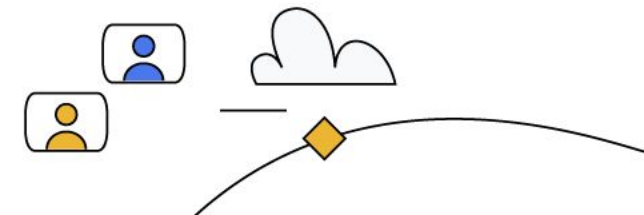
AI IMAGE TOOLS



AI AGENTS
AUTOMATING TASKS

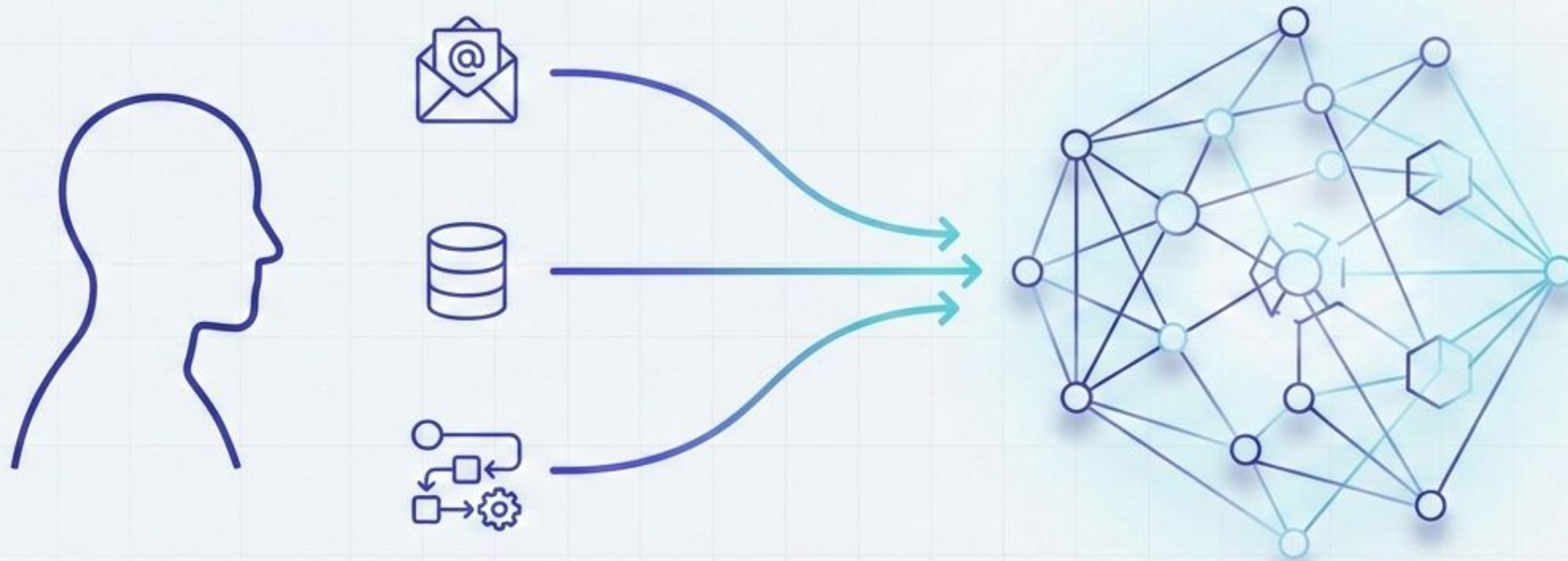


**WHAT THEY WILL
BUILD TODAY**



Why AI Matters — 2026 Reality: Automation + Agents

From manual work to autonomous systems



By 3PM
you will run your own AI
model and automate a
workflow



AI vs ML vs DL vs GAI

AI vs ML vs DL vs GAI

- ◆ **Artificial Intelligence (AI)**

Machines performing tasks that require human intelligence.
(The big umbrella.)

- ◆ **Machine Learning (ML)**

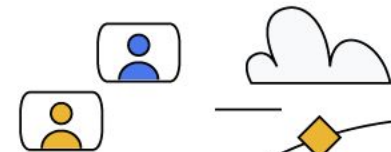
A subset of AI where systems **learn from data** instead of being programmed.

- ◆ **Deep Learning (DL)**

A subset of ML using **neural networks with many layers**.
Used in vision, speech, and large language models.

- ◆ **Generative AI**

A subset of Deep Learning that **creates new content** — text, images, code.
Example: ChatGPT



Artificial Intelligence

Is the field of study

Machine Learning

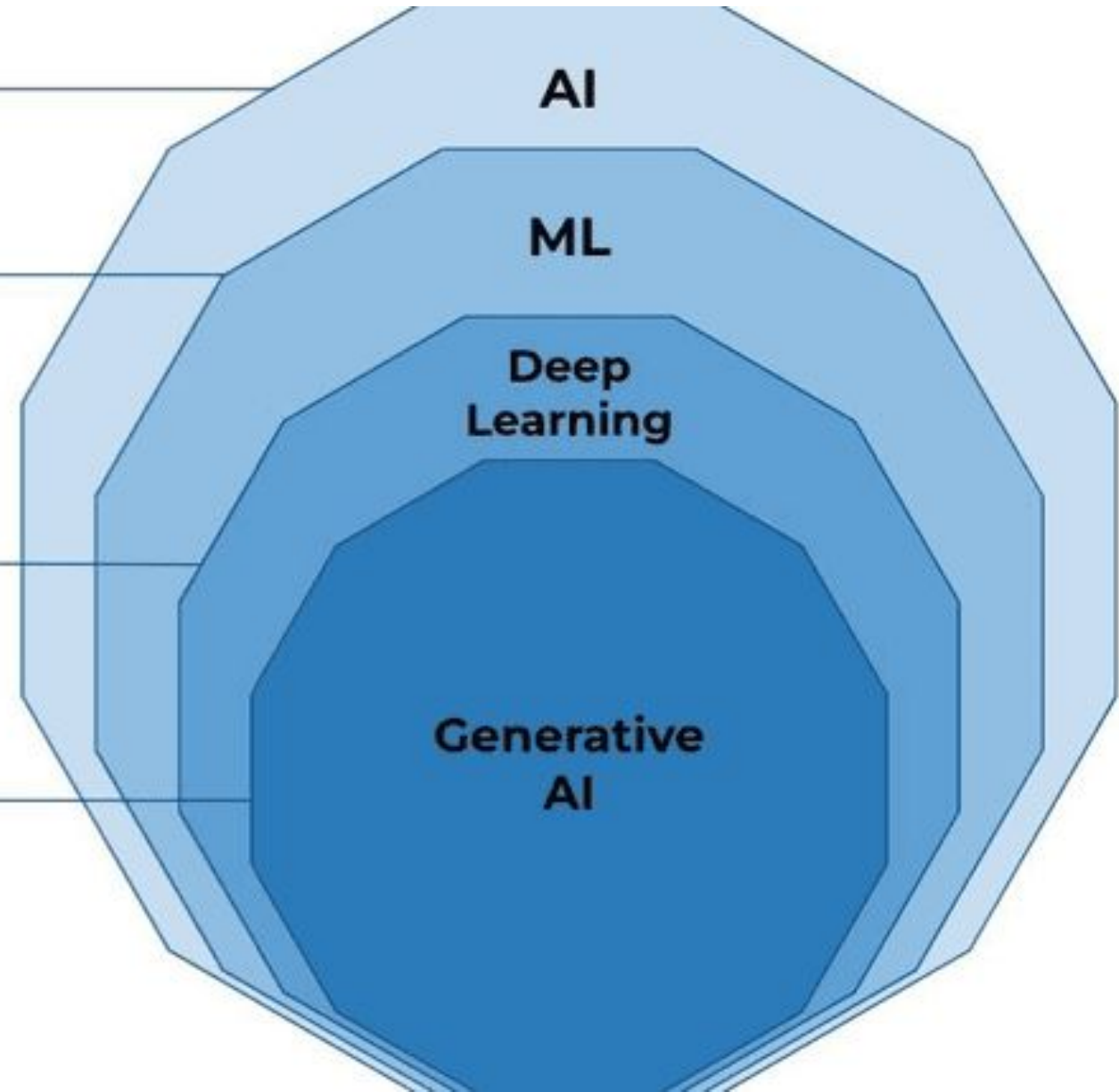
Is a branch of AI that focus on the creation of intelligent machines that learn from data. Another very well know branch inside AI is **Optimization**.

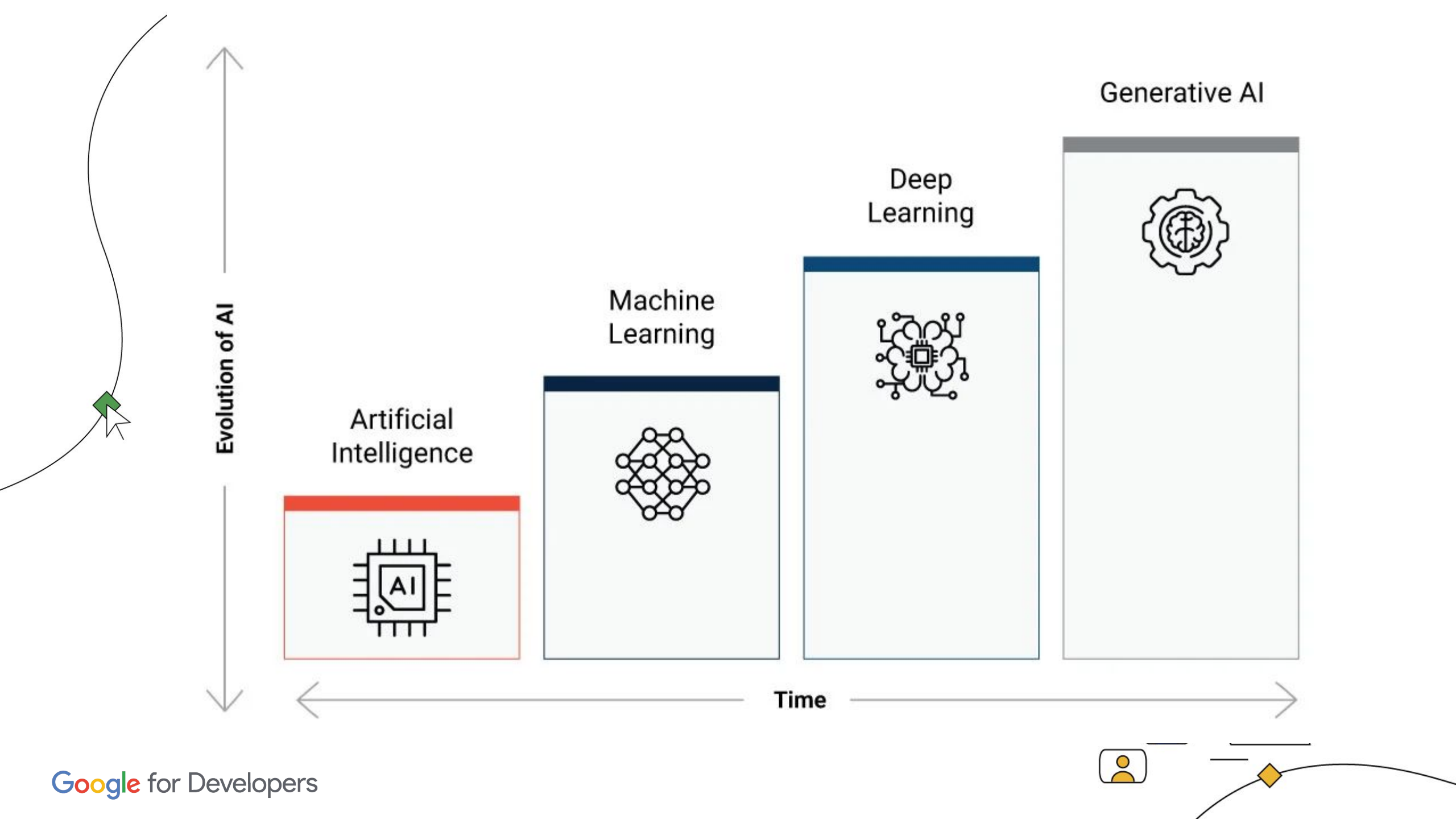
Deep Learning

Is a subset of Machine Learning methods, based on **Artificial Neural Networks**.
Examples: CNNs, RNNs

Generative AI

A type of ANNs that generate data that is similar to the data it was trained on.
Examples: GANs, LLMs





what is generative ai?

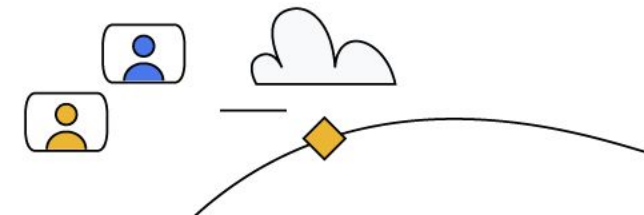
WHAT IS GENERATIVE AI?

Generative AI, or generative artificial intelligence, refers to machine learning models that can generate new content based on data they have already seen, when prompted.



WHAT IS GENERATIVE AI?

One model might generate text in response to a user's questions.



Generative Models

Generative **language** models

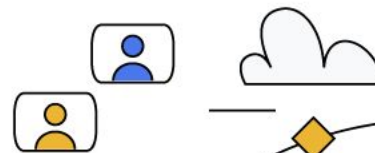
Generative language models learn about patterns in language through training data.

Then, given some text, they predict what comes next.

Generative **image** models

Generative image models produce new images using techniques like diffusion.

Then, given a prompt or related imagery, they transform random noise into images or generate images from prompts.



WHAT IS GENERATIVE AI?

Another can create videos for filmmakers to visualize their ideas using descriptive prompts.



(a) A dog wearing a superhero outfit with red cape flying through the sky.



WHAT IS GENERATIVE AI?

And another can even write code to help developers work faster.

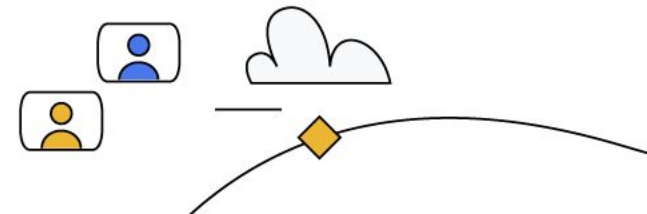
Uncaught ReferenceError: decomp is not defined

Debug with Ghostwriter

It seems like the problem is that the `decomp` library is not available in the global scope. Here's a fix:

```
(function (decomp) {  
  let points = font.textToPoints(letter, x, y,  
    sampleFactor: 0.1,  
    simplifyThreshold: 0,  
  );  
})
```

JavaScript

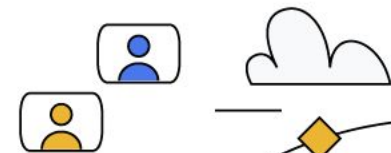
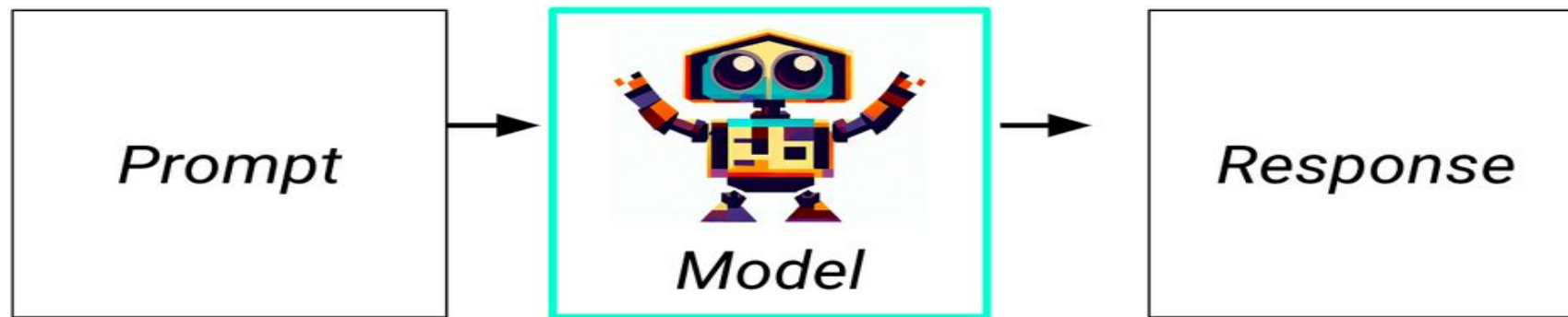


HOW DOES IT WORK?

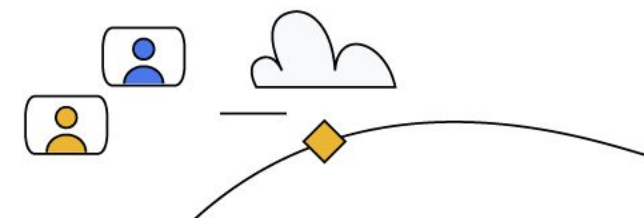
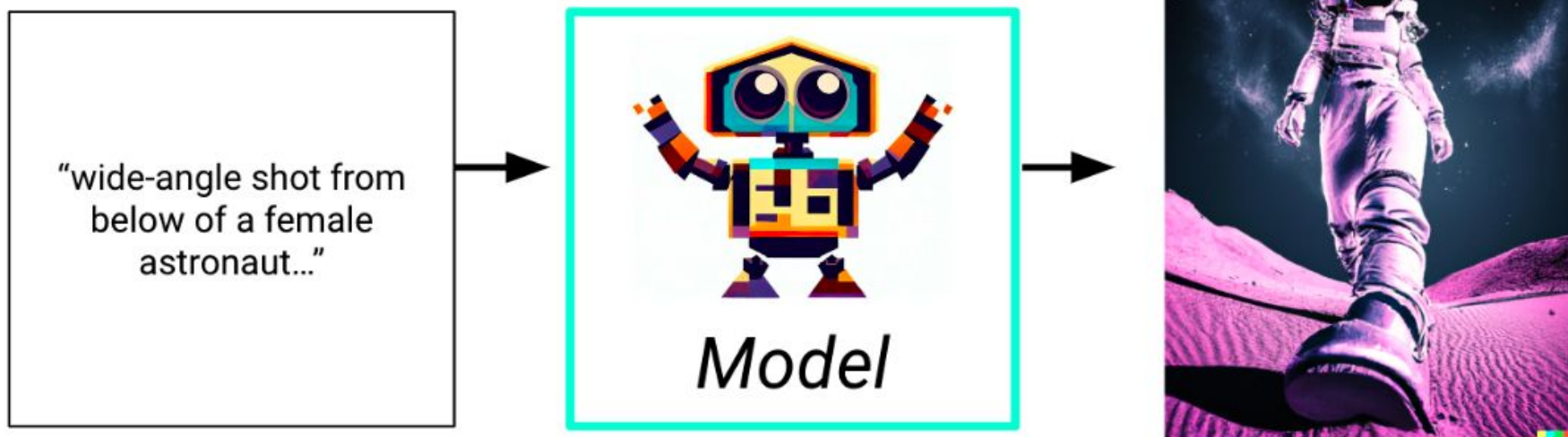
A generative AI model takes a **prompt** as input.

A prompt is a piece of data that guides the model to complete its task.

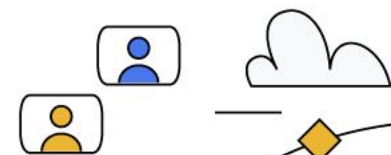
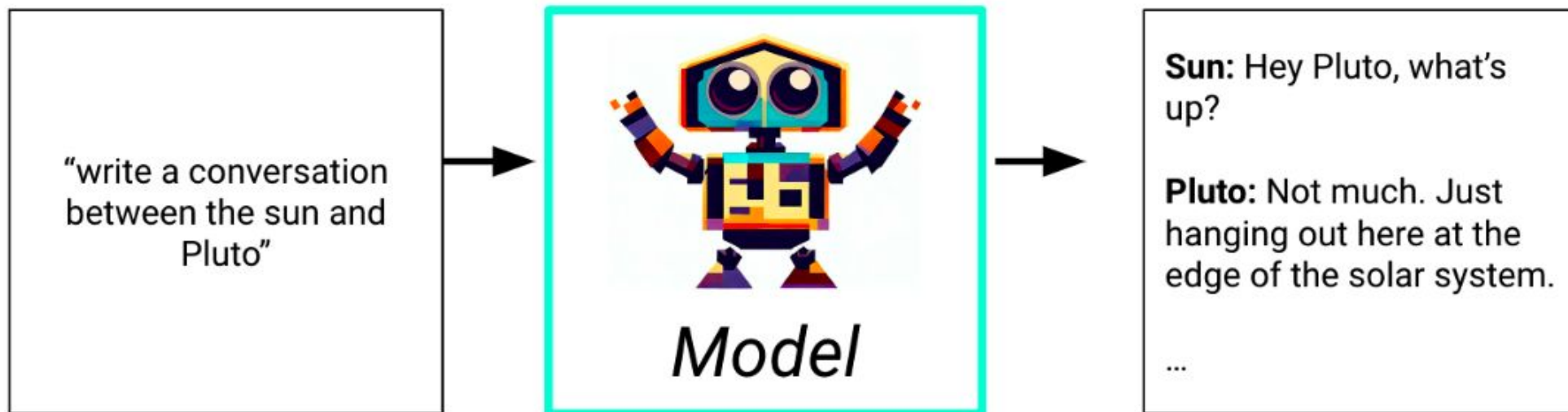
The model then processes the prompt to output, or generate, a response similar to data it has already seen.



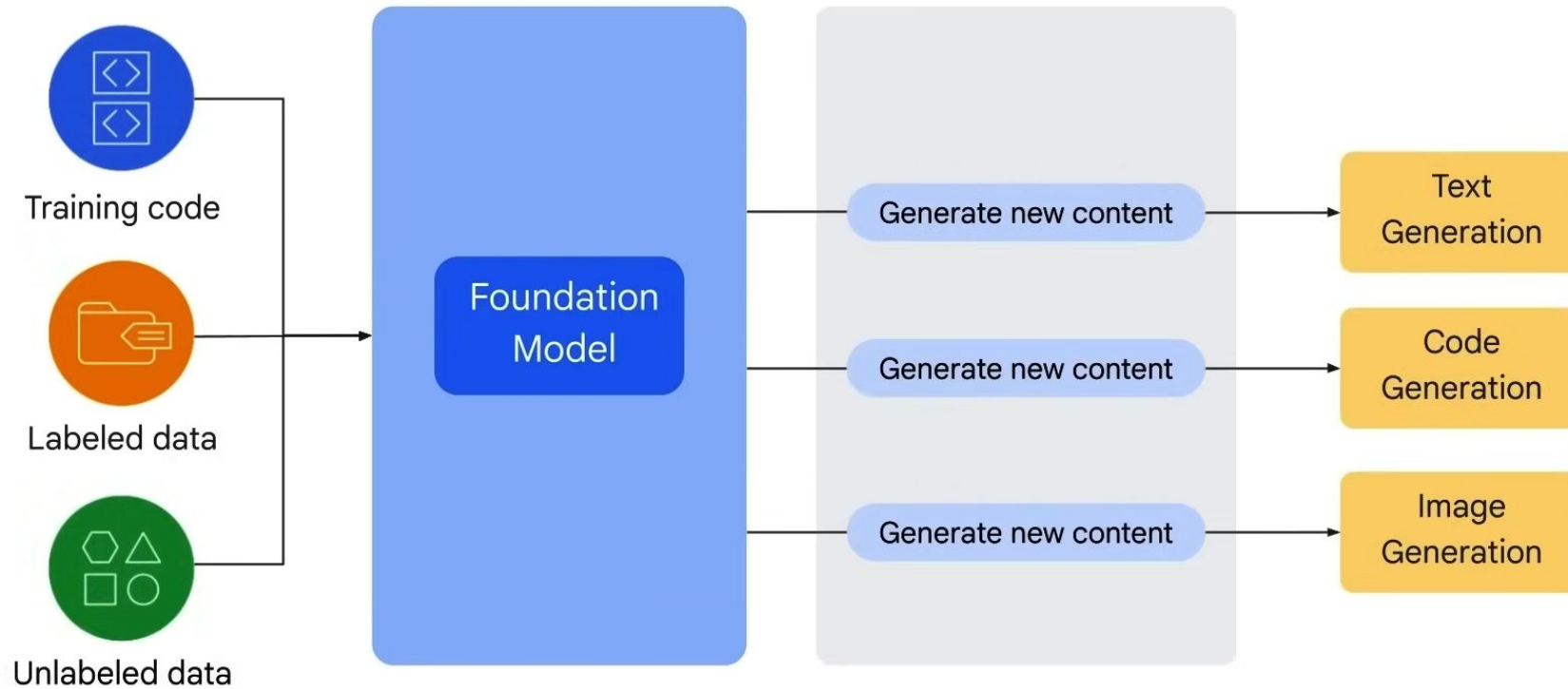
Create images



Hold conversations

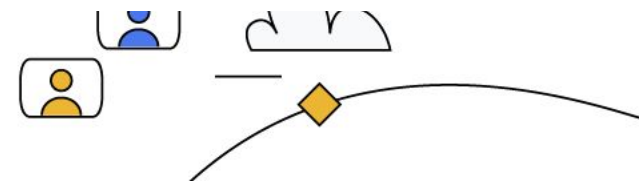
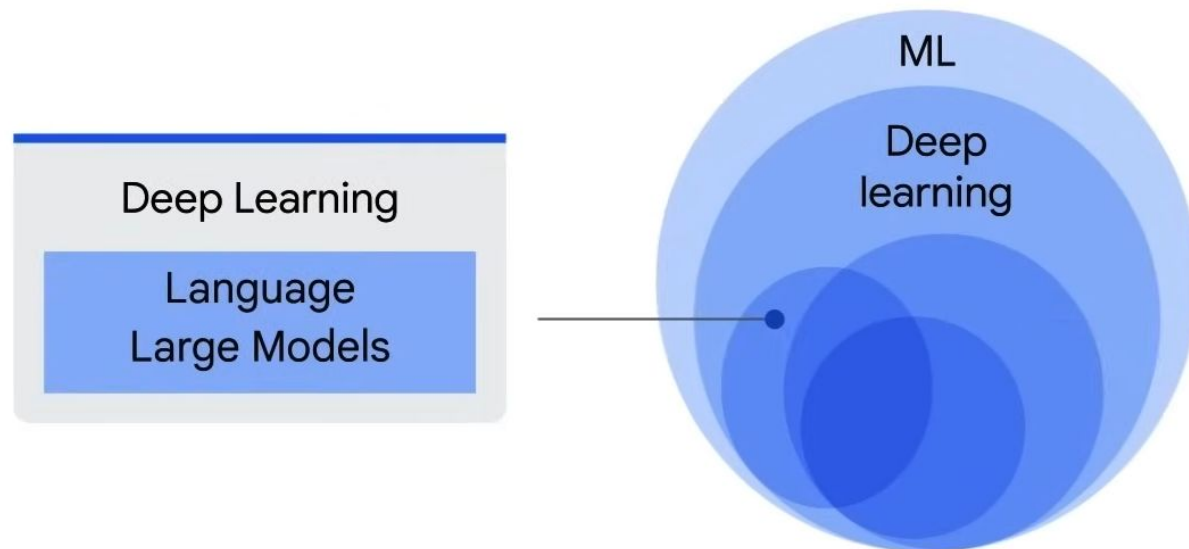


Gen AI Supervised, Semi-Supervised & Unsupervised Learning

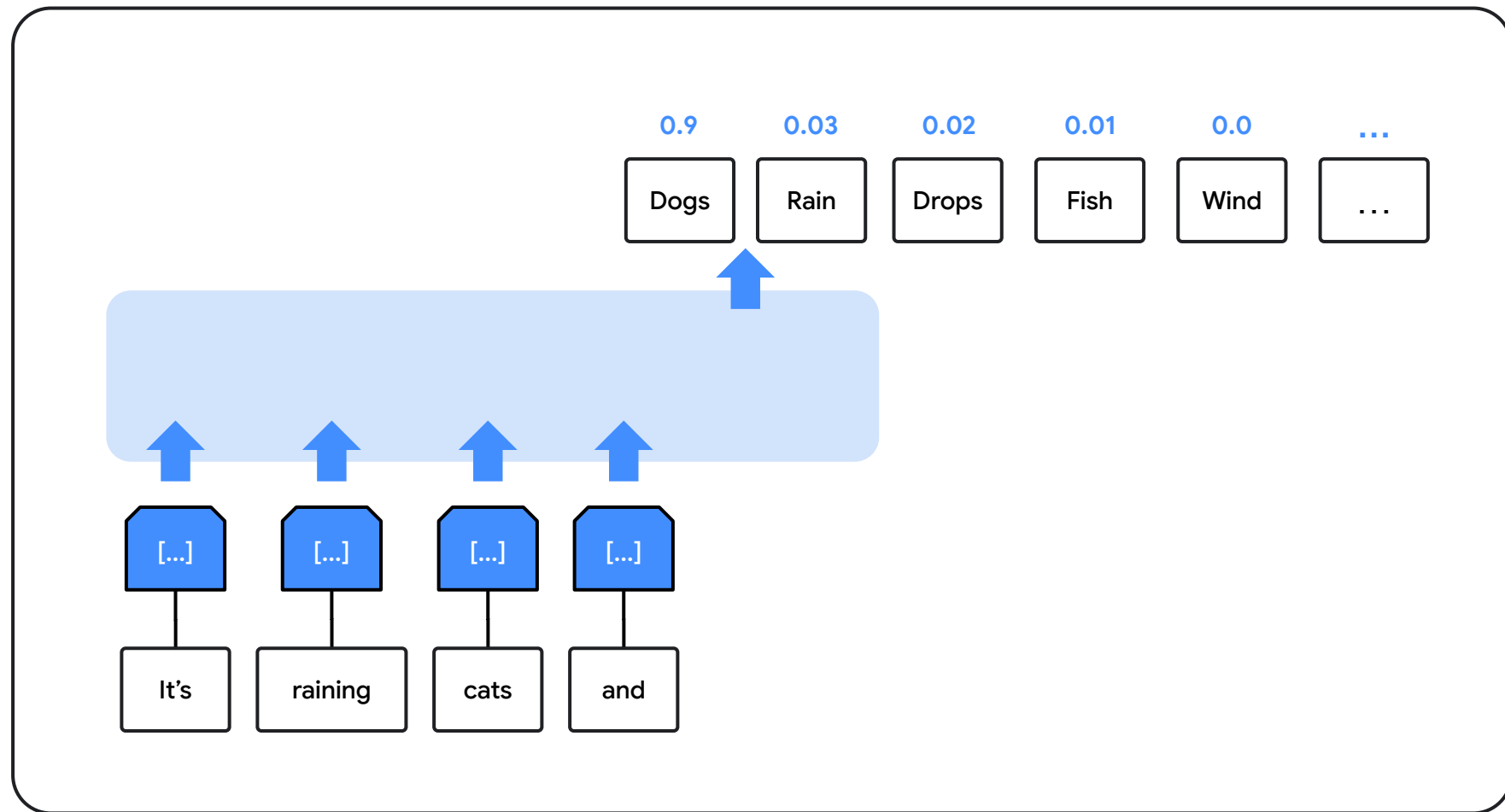




Large Language
Models (LLMs)
are also a **subset
of Deep Learning**



What is an LLM?



Classic Natural Language Problems



Entity extraction



Classification



Summarization



Sentiment Analysis



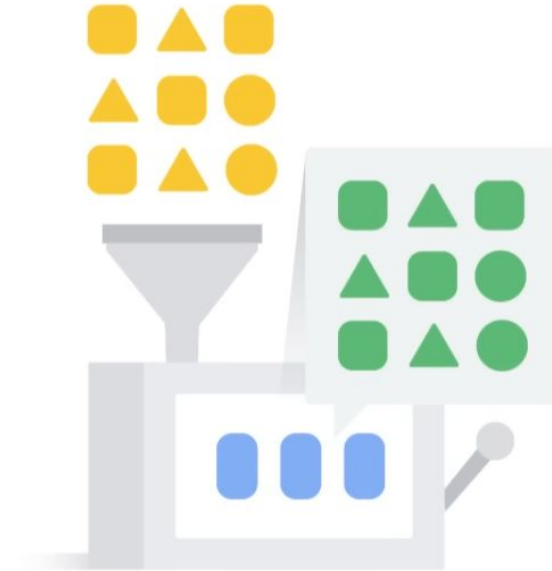
Translation



...

Large language models

- ✓ Large
 - Large training dataset
 - Large number of parameters
- ✓ General purpose
 - Commonality of human languages
 - Resource restriction
- ✓ Pre-trained and fine-tuned



LLMs let us
prototype *fast*.

User:

Explain this joke: “I used to be addicted to soap, but I'm clean now.”

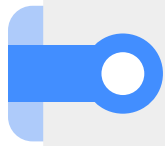
Model:

Model: The joke is a pun on the word "clean". It has two meanings:

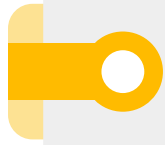
To be free of dirt or other unwanted substances.
To be free of addiction.

The joke is funny because it is unexpected. The listener expects the speaker to be talking about the first meaning of "clean", but they are actually talking about the second meaning. This creates a surprise and a laugh.

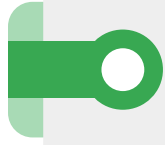
Why are large language models different?



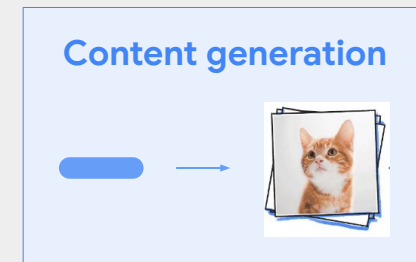
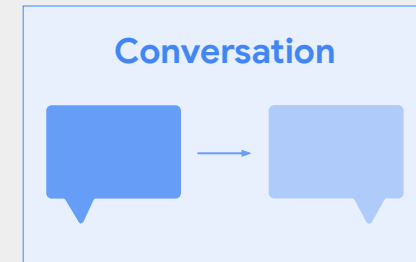
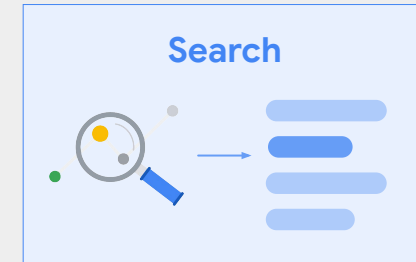
LLMs are characterized by **emergent abilities**, or the ability to perform tasks that were not present in smaller models.



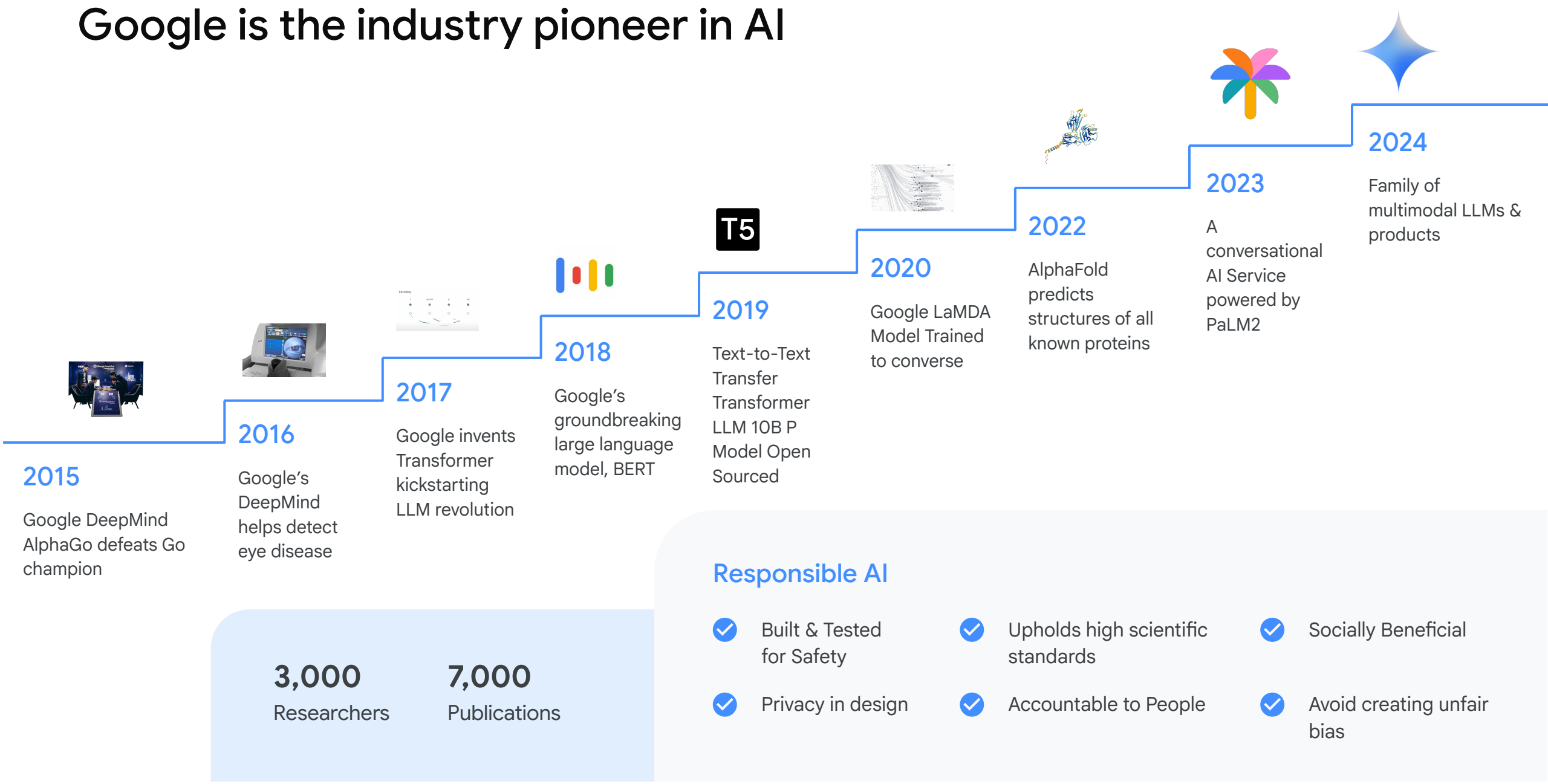
LLMs contextual understanding of human language **changes how we interact** with data and intelligent systems.



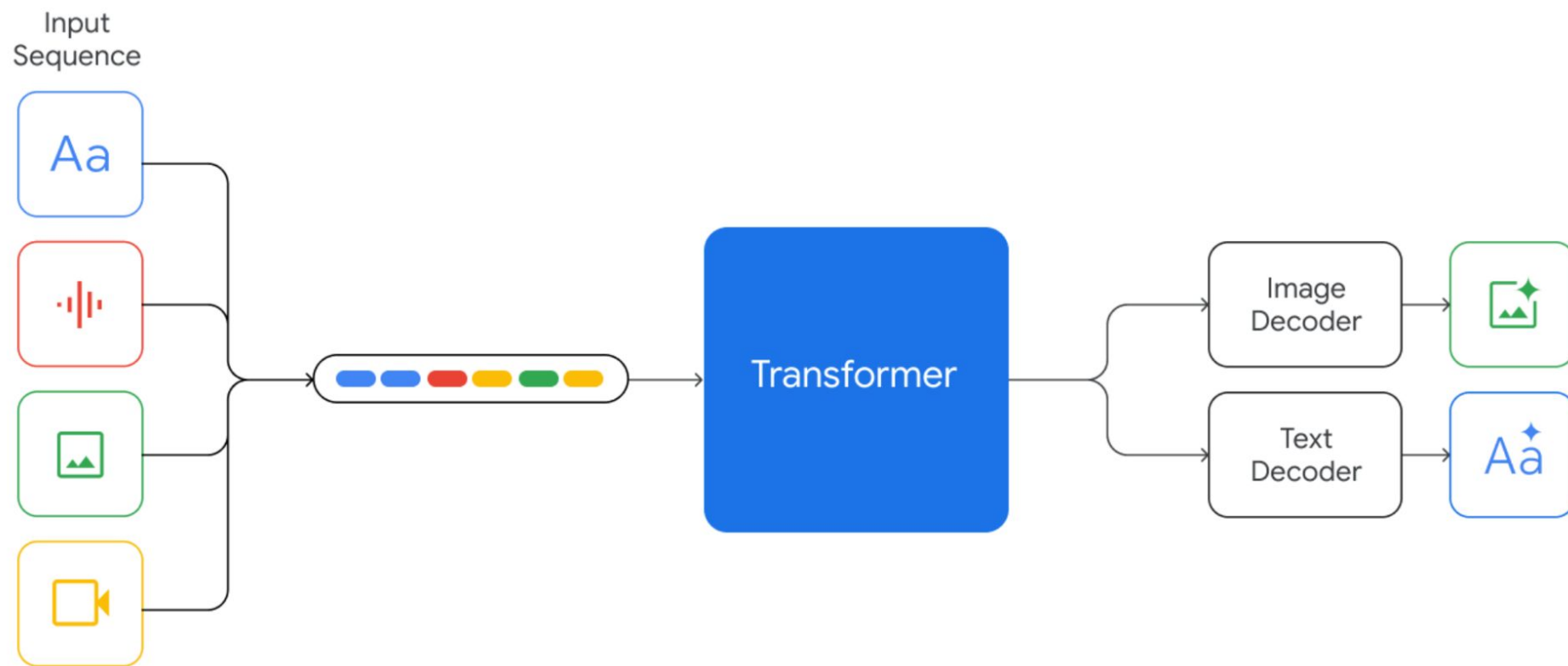
LLMs can find patterns and connections in **massive, disparate data corpora**.



Google is the industry pioneer in AI



Multimodality

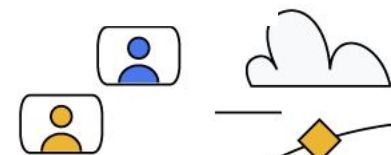


AI Ecosystem Overview



Landscape overview:

Category	Examples
Open Models	LLaMA, Mistral
Closed APIs	OpenAI, Claude
Local Serving	Ollama, vLLM
Automation	n8n
Deployment	Hugging Face Spaces
Hardware	NVIDIA GPUs



OPEN-SOURCE, LOCAL MODELS & NVIDIA

Why Open-Source Matters

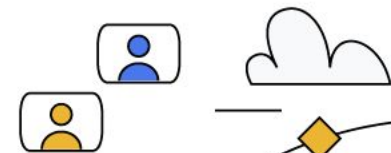
- Transparency & trust
- Community innovation
- Faster research progress
- Lower cost & accessibility

Why Local Models Matter

- Full data privacy & No API dependency
- Lower long-term cost
- Works offline

Where NVIDIA Fits

NVIDIA builds GPUs that power AI training and inference.
Modern models are trained on thousands of GPUs running in parallel.



Best Part !

PRACTICAL / HANDS-ON



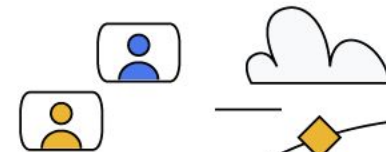
Run Local Model with
Ollama



WHAT IS OLLAMA?

Ollama is a lightweight tool that lets you **run large language models locally** on your computer with a simple command

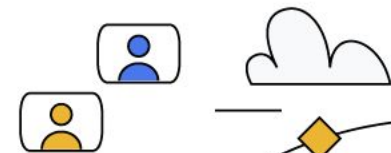
```
ollama run llama3
```





WHY USE OLLAMA?

- Run LLMs without cloud APIs
- Simple installation & setup
- Works with models like LLaMA, Mistral, Gemma
- Great for demos, testing, and private projects





BENEFITS OF OLLAMA

-  Full data privacy (runs locally)
-  No API cost
-  Fast local inference (with good GPU)
-  Easy model management

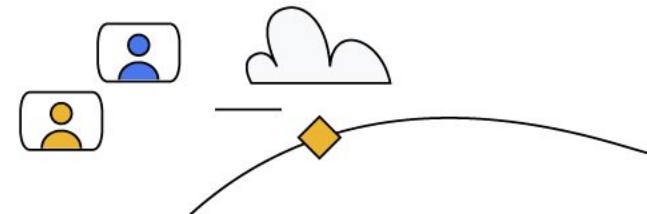


Hugging Face + vLLM Demo

WHAT IS HUGGING FACE?

Hugging Face

Hugging Face is an AI platform and open-source company that provides **pretrained machine learning models, datasets, and tools.**





WHAT IT OFFERS



Model Hub (100,000+ models)



Datasets library

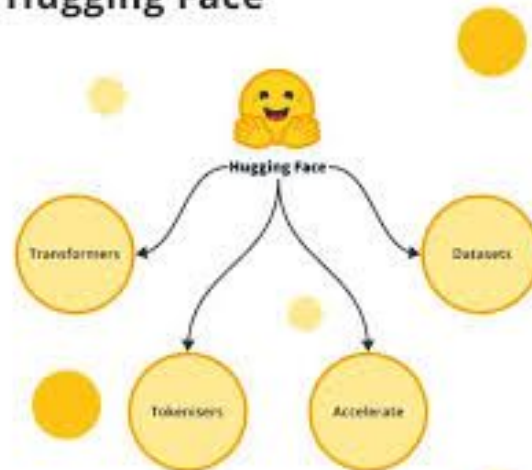


Transformers library (for NLP & LLMs)



Spaces (deploy AI demos easily)

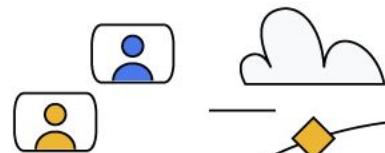
Hugging Face



WHAT IS VLLM? (HIGH-THROUGHPUT INFERENCE ENGINE)

⚡ **vLLM** is a high-throughput and memory-efficient inference engine for running Large Language Models (LLMs).

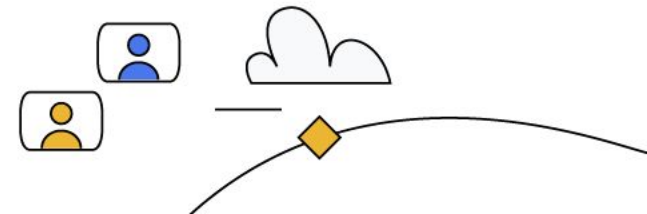
It is designed to serve LLMs **fast and at scale**.



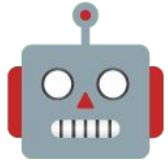


WHY IT'S POWERFUL

- High request throughput
- Efficient GPU memory usage
- Optimized for LLM serving
- Supports OpenAI-compatible APIs



AI Automation with n8n

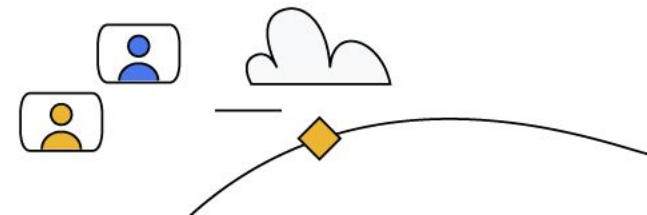


WHAT IS N8N?

n8n is an **open-source workflow automation tool** that connects apps, APIs, and AI to automate tasks without coding.



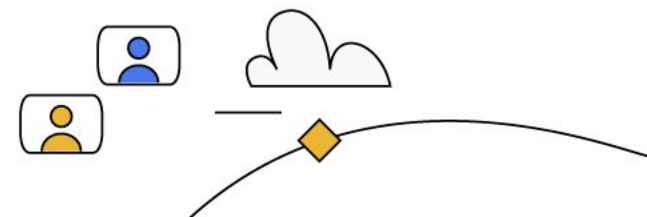
n8n





WHY USE N8N FOR AI AUTOMATION

- Automate repetitive tasks
- Connect AI models to your workflows
- Integrate multiple apps & APIs easily
- Visual drag-and-drop workflow editor





BENEFITS



Save time & reduce errors



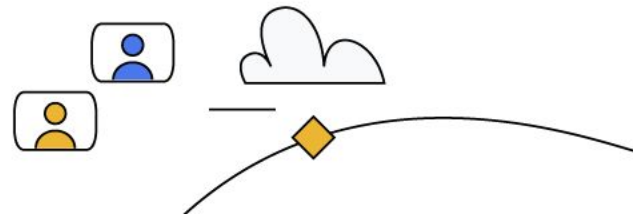
Keep data in control (self-hosted)

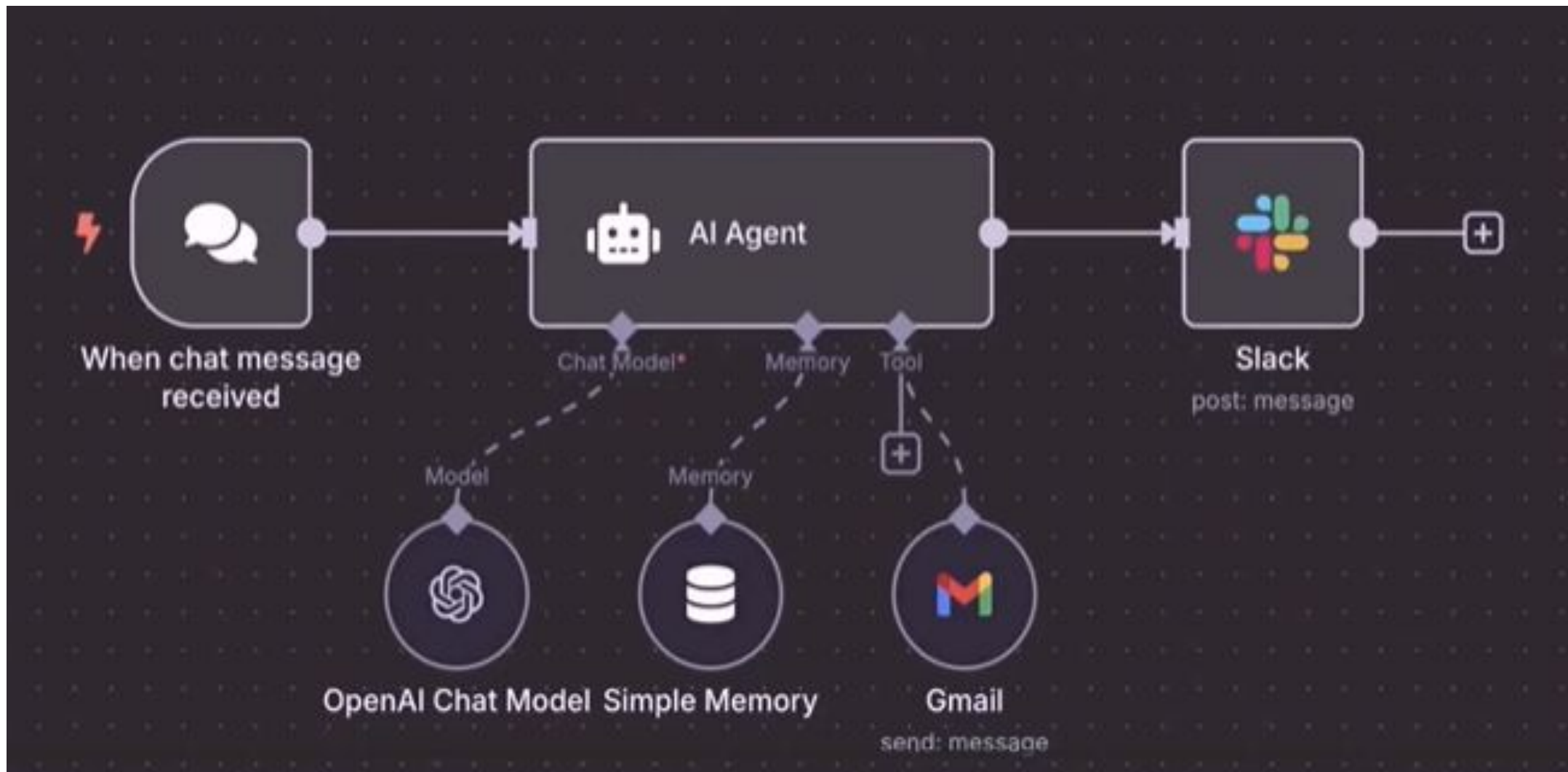


Build smart AI-powered workflows quickly



n8n







Thank you!



Imen Selmi

WTM, GDG Sfax

